

Practical Task 13: Ethical Hacking

Question 1:

- a. **Nmap** is the correct tool for mapping network topology as it operates at OSI layers 3 and 4 to discover active hosts, open ports and running services.
- b.
 - i. There are 3 active hosts
 - ii. The port scan of 10.0.2.15 revealed no open ports. All 1000 scanned TCP ports returned a filtered (no response) state.
 - iii. The -sV service detection scan confirmed no services were detectable on 10.0.2.15, consistent with all ports being in a filtered state.
- c. There are 3 devices on my local network

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(kali@kali)-[~]
$ nmap -sn 10.0.2.15/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 05:52 -0400
Nmap scan report for 10.0.2.2
Host is up (0.00059s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.3
Host is up (0.00088s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.74 seconds

(kali@kali)-[~]
$ nmap 10.0.2.15
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 05:53 -0400
Stats: 0:00:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 4.00% done; ETC: 05:57 (0:03:36 remaining)
Nmap scan report for 10.0.2.15
Host is up.
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 206.63 seconds

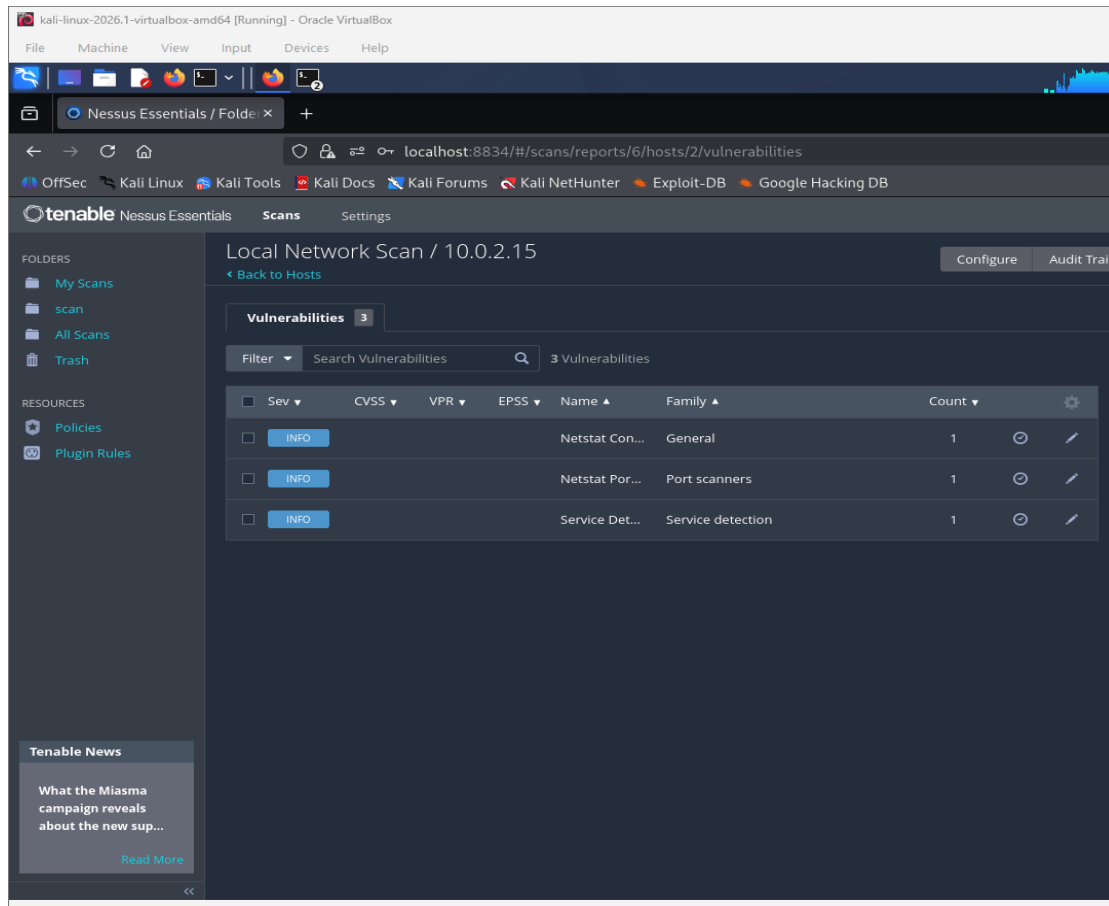
(kali@kali)-[~]
$ nmap -sV 10.0.2.15
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 05:57 -0400
Stats: 0:00:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.50% done
Nmap scan report for 10.0.2.15
Host is up.
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

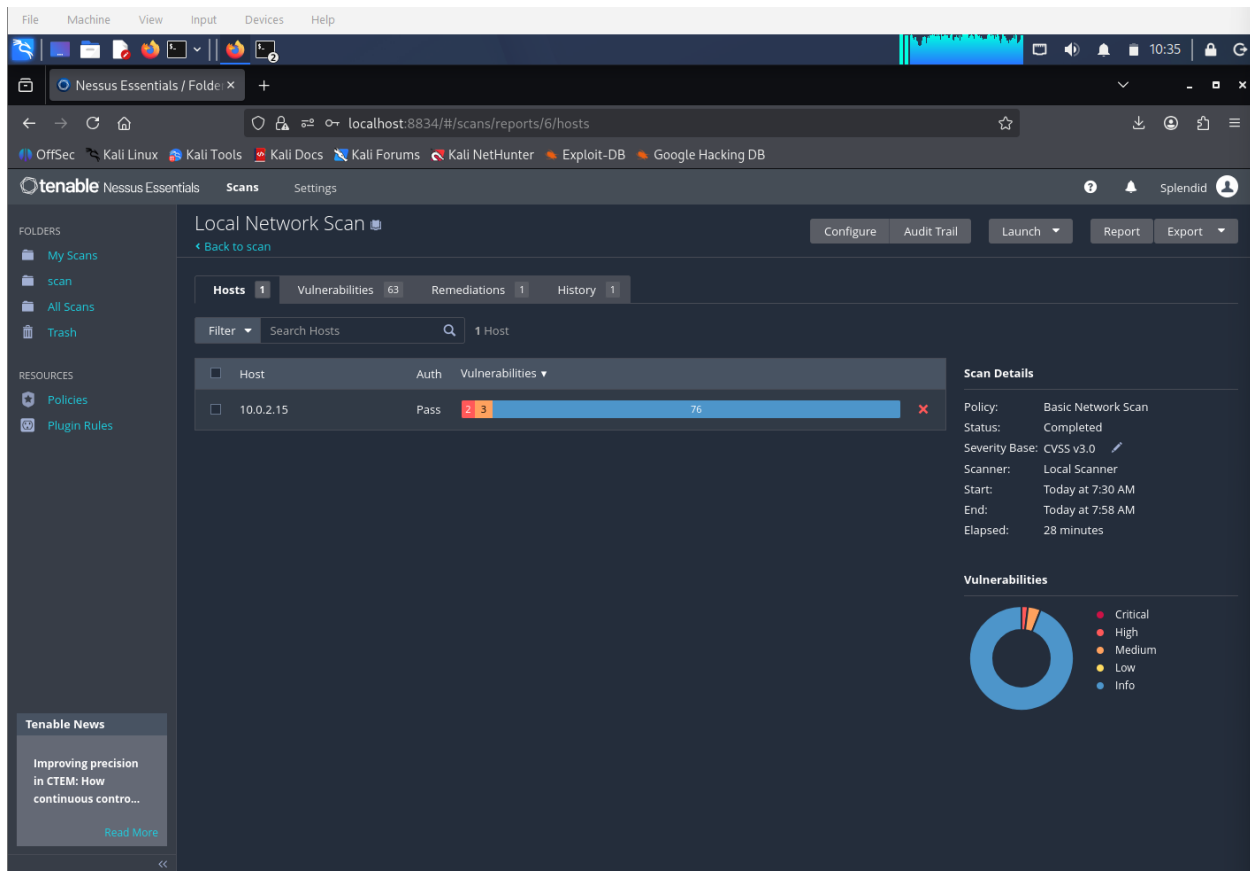
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 207.26 seconds

(kali@kali)-[~]
$
```

Question 2:

- a. **Nessus** is the correct tool for vulnerability scanning as it operates at OSI layers 6 and 7 providing detailed vulnerability assessment including missing patches, misconfigurations and compliance gaps.
- b.





c. The scan of host 10.0.2.15 returned a total of 3 findings, all classified at the **INFO** severity level:

- Critical: 0
- High: 0
- Medium: 0
- Low: 0
- Info: 3

The 3 **INFO** findings identified were Netstat Connections, Netstat Ports, and Service Detection. Nessus identifies severity levels through color-coded badges:

- Red indicates Critical
- Orange indicates High
- Yellow indicates Medium

- Green indicates Low
- Light Blue indicates Info.

The donut chart within the scan results visually confirmed that only Info-level findings were present, indicating that no exploitable vulnerabilities were detected on the local network during this unauthenticated scan.